

Pełczyński V^* -Type Properties of Free Locally Convex Spaces: A Complete Classification

Full solution packet, human-review draft

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Abstract

Let $L(X)$ be the free locally convex space over a Tychonoff space X , and let $1 \leq p \leq q \leq \infty$. We give a complete answer to Problem 3.18 in Gabrielyan's paper on Gelfand–Phillips type properties[5]. The properties $V_{(p,q)}^*$, $EV_{(p,q)}^*$, $sV_{(p,q)}^*$, and $sEV_{(p,q)}^*$ are all equivalent to the condition that every functionally bounded subset of X is finite. In contrast, $L(X)$ has $wsV_{(p,q)}^*$ for every Tychonoff X , and it has $wsEV_{(p,q)}^*$ for every X whenever $q < \infty$. At the sole exceptional endpoint $q = \infty$, the class of p -(EV^*) sets is exactly the class of bounded sets. Consequently, $wsEV_p^*$ is equivalent to the weak Cauchy subsequence property of $L(X)$. We translate this last condition into an exact sequential compactness condition for finitely supported probability measures on compact closures of countable functionally bounded subsets of X in the Dieudonné completion.

Source problem evidence

and hence relatively compact. Thus $L(X)$ trivially has the coarse GP_p property. □

Theorems 3.16 and 3.17 suggest the following problem.

Problem 3.18. *Let $1 \leq p \leq q \leq \infty$. Characterize Tychonoff spaces X for which $L(X)$ has one of the $V_{(p,q)}^*$ type properties defined in Definition 2.5.*

We know from (ii) of Lemma 3.1 that the case $(1, \infty)$ is the strongest one in the sense that if E

The source problem is Problem 3.18 in Gabrielyan's paper [5]. It asks: for $1 \leq p \leq q \leq \infty$, characterize the Tychonoff spaces X for which $L(X)$ has one of the $V_{(p,q)}^*$ -type properties from Definition 2.5 of that paper. The definition includes $V_{(p,q)}^*$, $EV_{(p,q)}^*$, $sV_{(p,q)}^*$, $sEV_{(p,q)}^*$, $wsV_{(p,q)}^*$, and $wsEV_{(p,q)}^*$.

1 Introduction and the classification theorem

All locally convex spaces are Hausdorff and are over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. For a Tychonoff space X , every element of the free locally convex space $L(X)$ has a unique representation

$$\chi = \sum_{j=1}^m a_j x_j,$$

where the x_j are distinct points of X . We write

$$\|\chi\|_1 = \sum_{j=1}^m |a_j|, \quad \text{supp}(\chi) = \{x_1, \dots, x_m\}.$$

For $A \subseteq L(X)$, set

$$\text{supp}(A) = \bigcup_{\chi \in A} \text{supp}(\chi), \quad C_A = \sup_{\chi \in A} \|\chi\|_1.$$

The dual of $L(X)$ is $C(X)$, with the canonical pairing

$$\langle f, \chi \rangle = \sum_{j=1}^m a_j f(x_j).$$

We recall the definitions needed for the statement. A nonempty set A in a locally convex space E is a (p, q) -(V^*) set if, for every weakly p -summable sequence (u_n) in the strong dual E'_β , the sequence

$$\left(\sup_{a \in A} |\langle u_n, a \rangle| \right)_n$$

belongs to ℓ_q when $q < \infty$, and belongs to c_0 when $q = \infty$. A (p, q) -(EV^*) set is defined in the same way, but only equicontinuous weakly p -summable sequences are tested. The properties $V_{(p,q)}^*$ and $EV_{(p,q)}^*$ require these sets to be relatively weakly compact; the properties $sV_{(p,q)}^*$ and $sEV_{(p,q)}^*$ require relative weak sequential compactness; and $wsV_{(p,q)}^*$ and $wsEV_{(p,q)}^*$ require weak sequential precompactness, meaning that every sequence has a weakly Cauchy subsequence. When $q = \infty$, the second subscript is omitted.

The following theorem is the full classification proved below.

Theorem 1.1 (Complete classification). *Let X be a Tychonoff space and let $1 \leq p \leq q \leq \infty$.*

(I) *The following assertions are equivalent:*

- (a) *every functionally bounded subset of X is finite;*
- (b) *$L(X)$ has $V_{(p,q)}^*$;*
- (c) *$L(X)$ has $EV_{(p,q)}^*$;*
- (d) *$L(X)$ has $sV_{(p,q)}^*$;*
- (e) *$L(X)$ has $sEV_{(p,q)}^*$.*

(II) *The space $L(X)$ has $wsV_{(p,q)}^*$ for every Tychonoff space X .*

(III) *If $q < \infty$, then $L(X)$ has $wsEV_{(p,q)}^*$ for every Tychonoff space X .*

(IV) *For $q = \infty$, the following assertions are equivalent, and do not depend on p :*

- (a) *$L(X)$ has $wsEV_p^*$;*
- (b) *every bounded sequence in $L(X)$ has a weakly Cauchy subsequence;*
- (c) *X has the atomic sequential compactness property ASCP of Definition 6.3 below.*

Thus only the equicontinuous endpoint $wsEV_{(p,\infty)}^*$ depends on X in a nontrivial way. The condition in Theorem 1.1(IV) will be expressed solely in terms of probability measures over compact closures of countable functionally bounded subsets of X .

Proof intuition

The four compactness and weak-sequential-compactness properties are governed by one simple obstruction: an infinite functionally bounded set in X supports weighted finite partial sums in $L(X)$ which are V^* -small but converge pointwise to an infinite measure outside $L(X)$. This proves the necessity of “functionally bounded sets are finite”, while the converse follows because bounded EV^* -sets then live in finite-dimensional coordinate spans.

The weak sequentially precompact variants require a different mechanism. If a bounded subset of $L(X)$ contains no weakly Cauchy subsequence, its supports sit in a compact $K \subseteq \mu X$, and the corresponding measures fail weak compactness in $M(K)$. The Dieudonné–Grothendieck disjoint-open-set criterion gives disjoint tests on K . A telescoping extension turns these tests into a weakly 1-summable sequence in $C(X)_\beta$ which detects the failure uniformly. This immediately forces every V^* -set to be weakly sequentially precompact and, after a c_0 -damping trick, gives the equicontinuous result for finite q . The remaining endpoint $wsEV_p^*$ is exactly the assertion that bounded sequences in $L(X)$ have weakly Cauchy subsequences; this is reformulated as a sequential compactness condition for finitely supported probability measures on compact closures of countable functionally bounded subsets of X .

2 Free locally convex preliminaries

Let μX denote the Dieudonné completion of X . We use the following standard bounded-set criterion for free locally convex spaces [4, Proposition 2.7].

Proposition 2.1 (Bounded sets in $L(X)$). *For a nonempty set $B \subseteq L(X)$, the following are equivalent:*

- (i) B is bounded in $L(X)$;
- (ii) $\text{supp}(B)$ has compact closure in μX and $C_B < \infty$;
- (iii) $\text{supp}(B)$ is functionally bounded in X and $C_B < \infty$.

We record several consequences in a form used repeatedly below.

Lemma 2.2 (Restriction to compact closures). *Let $D \subseteq X$ be functionally bounded and put*

$$K_D = \text{cl}_{\mu X} D.$$

Then K_D is compact. Every $f \in C(X)$ has a unique continuous extension $\bar{f} \in C(\mu X)$, and the restriction operator

$$R_D : C(X) \longrightarrow C(K_D), \quad R_D f = \bar{f}|_{K_D},$$

is onto. Moreover,

$$\|R_D f\|_\infty = \sup_{x \in D} |f(x)|.$$

Proof. A subset of X is functionally bounded if and only if its closure in μX is compact; this is part of the standard theory of the universal uniformity and is also used in Proposition 2.1. Every continuous scalar-valued function on X is uniformly continuous for the universal uniformity, hence extends uniquely to μX .

It remains to prove surjectivity. A compact subset of a Tychonoff space is C -embedded. Indeed, regard K_D as a closed subset of the Stone–Čech compactification $\beta(\mu X)$, extend a scalar-valued continuous function on K_D by the Tietze extension theorem, and restrict the extension to μX and then to X . Finally, D is dense in K_D , so the displayed norm identity holds. $\square$

Lemma 2.3 (The strong dual of $L(X)$). *The strong topology $\beta(C(X), L(X))$ is generated by the seminorms*

$$p_D(f) = \sup_{x \in D} |f(x)|,$$

where D ranges over the functionally bounded subsets of X . Furthermore, for every $\Lambda \in (C(X)_\beta)'$ there are a functionally bounded set $D \subseteq X$ and a finite regular Borel measure $\nu \in M(K_D) = C(K_D)'$ such that

$$\Lambda(f) = \int_{K_D} R_D f \, d\nu \quad (f \in C(X)).$$

Proof. If D is functionally bounded, then $\{x : x \in D\}$ is bounded in $L(X)$ by Proposition 2.1; hence p_D is a strong-dual seminorm. Conversely, if $B \subseteq L(X)$ is bounded, then

$$\sup_{\chi \in B} |\langle f, \chi \rangle| \leq C_B \sup_{x \in \text{supp}(B)} |f(x)|,$$

and $\text{supp}(B)$ is functionally bounded. This proves the first assertion.

If Λ is continuous, there are a functionally bounded D and $c > 0$ such that

$$|\Lambda(f)| \leq cp_D(f) \quad (f \in C(X));$$

a finite union absorbs the finitely many seminorms arising from a basic neighborhood. Thus Λ vanishes on $\ker R_D$. By Lemma 2.2, it induces a bounded linear functional on $C(K_D)$. The Riesz representation theorem gives the required measure ν . $\square$

We shall also use two standard Banach-space facts. First, by the Eberlein–Šmulian theorem, a relatively weakly compact subset of a Banach space is relatively weakly sequentially compact. Second, we use the following form of the Dieudonné–Grothendieck criterion [8].

Theorem 2.4 (Dieudonné–Grothendieck criterion). *Let K be compact and let $H \subseteq M(K)$ be norm bounded. Then H is relatively weakly compact if and only if, for every sequence (U_n) of pairwise disjoint open subsets of K ,*

$$\sup_{\mu \in H} |\mu(U_n)| \longrightarrow 0.$$

3 The four compactness and sequential-compactness variants

We begin with the universal obstruction supplied by an infinite functionally bounded set.

Lemma 3.1 (Weighted partial sums). *Let $D = \{x_i : i \in \mathbb{N}\}$ be an infinite functionally bounded subset of X , with all x_i distinct, and let $w_i = 2^{-i-1}$. Put*

$$a_m = \sum_{i=0}^m w_i x_i, \quad A = \{a_m : m \in \mathbb{N}\}.$$

Then A is a (p, q) -(V^) set in $L(X)$ for every $1 \leq p \leq q \leq \infty$.*

Proof. The restriction operator

$$R : C(X)_\beta \longrightarrow \ell_\infty, \quad Rf = (f(x_i))_{i \in \mathbb{N}},$$

is continuous by Lemma 2.3. Define

$$H : \ell_\infty \longrightarrow \ell_\infty, \quad H(y)(m) = \sum_{i=0}^m w_i y(i).$$

Assume first that $1 \leq p < \infty$ and let (y_k) be weakly p -summable in ℓ_∞ . The operator

$$S : (\ell_\infty)' \longrightarrow \ell_p, \quad S(\varphi) = (\varphi(y_k))_k,$$

is well defined and has closed graph, hence is bounded. If e'_i is the i th coordinate functional, then

$$\|(y_k(i))_k\|_{\ell_p} = \|S(e'_i)\|_{\ell_p} \leq \|S\|.$$

Minkowski's inequality therefore gives

$$\begin{aligned} \left(\sum_k \|H(y_k)\|_\infty^p \right)^{1/p} &\leq \sum_i w_i \left(\sum_k |y_k(i)|^p \right)^{1/p} \\ &\leq \|S\| \sum_i w_i < \infty. \end{aligned}$$

Thus $(\|H(y_k)\|_\infty)_k \in \ell_p$, and hence it belongs to ℓ_q when $q < \infty$ and to c_0 when $q = \infty$.

For $p = \infty$, the finite-rank operators

$$H_N(y)(m) = \sum_{i=0}^{\min\{m, N\}} w_i y(i)$$

converge to H in operator norm. Hence H is compact and sends weakly null sequences to norm-null sequences.

Now let (f_k) be weakly p -summable in $C(X)_\beta$. Then (Rf_k) is weakly p -summable in ℓ_∞ , and

$$\sup_{a \in A} |\langle f_k, a \rangle| = \sup_m \left| \sum_{i=0}^m w_i f_k(x_i) \right| = \|H(Rf_k)\|_\infty.$$

The preceding estimates prove the assertion. $\square$

Lemma 3.2 (The weighted partial sums do not compactify in $L(X)$). *With the notation of Lemma 3.1, the set A is neither relatively weakly compact nor relatively weakly sequentially compact in $L(X)$.*

Proof. Define

$$\mu(f) = \sum_{i=0}^{\infty} w_i f(x_i), \quad f \in C(X).$$

The series converges absolutely because D is functionally bounded. It also defines an element of $(C(X)_\beta)'$, since

$$|\mu(f)| \leq \left(\sum_i w_i \right) p_D(f).$$

For every $f \in C(X)$,

$$\langle f, a_m \rangle \longrightarrow \mu(f).$$

The functional μ is not represented by an element of $L(X)$. Otherwise, write $\mu = \sum_{j=1}^r c_j y_j$. Choose N with $x_N \notin \{y_1, \dots, y_r\}$. Complete regularity supplies $f \in C(X, [0, 1])$ such that $f(x_N) = 1$ and f vanishes on

$$\{y_1, \dots, y_r, x_0, \dots, x_{N-1}\}.$$

The finite representation gives $\mu(f) = 0$, whereas the series gives $\mu(f) \geq w_N > 0$, a contradiction.

Let

$$J : L(X)_w \longrightarrow \mathbb{F}^{C(X)}, \quad J(\chi) = (\langle f, \chi \rangle)_{f \in C(X)},$$

be the canonical weak-topology embedding. If the weak closure of A were compact, its image under J would be compact and therefore closed in the Hausdorff product $\mathbb{F}^{C(X)}$. Since $J(a_m)$ converges coordinatewise to $(\mu(f))_{f \in C(X)}$, that point would have to belong to $J(L(X))$, contrary to the preceding paragraph. Thus A is not relatively weakly compact. Every subsequence of (a_m) has the same coordinatewise limit μ , so no subsequence converges weakly in $L(X)$. Hence A is not relatively weakly sequentially compact. $\square$

We can now settle the first four properties.

Theorem 3.3. *Let $1 \leq p \leq q \leq \infty$. The following are equivalent:*

- (i) *every functionally bounded subset of X is finite;*
- (ii) *$L(X)$ has $V_{(p,q)}^*$;*
- (iii) *$L(X)$ has $\text{EV}_{(p,q)}^*$;*
- (iv) *$L(X)$ has $sV_{(p,q)}^*$;*
- (v) *$L(X)$ has $s\text{EV}_{(p,q)}^*$.*

Proof. Assume first that every functionally bounded subset of X is finite. Every (p, q) -(V^*) set is a (p, q) -(EV^*) set, and every (p, q) -(EV^*) set is bounded [6, Lemma 7.2]. If B is such a set, then Proposition 2.1 gives $C_B < \infty$ and says that $F = \text{supp}(B)$ is functionally bounded. Thus F is finite and

$$B \subseteq \left\{ \sum_{x \in F} a_x x : \sum_{x \in F} |a_x| \leq C_B \right\}.$$

The set on the right is compact in the finite-dimensional space $\text{span}(F)$. Hence B is relatively compact, relatively weakly compact, relatively weakly sequentially compact, and weakly sequentially precompact. This proves all positive implications.

Conversely, if X contains an infinite functionally bounded subset, Lemmas 3.1 and 3.2 provide a (p, q) -(V^*) set which is neither relatively weakly compact nor relatively weakly sequentially compact. Since every (V^*) set is an (EV^*) set, the same witness defeats all four properties. $\square$

4 A detector for failure of weak sequential precompactness

The new ingredient needed for the weak variants is the following lemma. It turns a bounded sequence without a weakly Cauchy subsequence into a weakly 1-summable sequence of continuous functions that detects that failure uniformly.

Lemma 4.1 (Disjointization and telescoping detector). *Let $A \subseteq L(X)$ be bounded and suppose that A contains a sequence with no weakly Cauchy subsequence. Then there are $\eta > 0$ and functions $h_n \in C(X, [0, 1])$ such that*

- (i) *(h_n) is weakly 1-summable in $C(X)_\beta$;*
- (ii) $\sum_{n=0}^{\infty} h_n(x) \leq 1$ *for every $x \in X$;*
- (iii) $\sup_{\chi \in A} |\langle h_n, \chi \rangle| \geq \eta$ *for every n .*

Proof. Choose a sequence $(\chi_m) \subseteq A$ with no weakly Cauchy subsequence and put

$$S = \bigcup_m \text{supp}(\chi_m), \quad K = \text{cl}_{\mu_X} S.$$

The set S is countable and functionally bounded, and the total variations $\|\chi_m\|_1$ are uniformly bounded. We therefore regard the χ_m as a bounded sequence in $M(K) = C(K)'$.

The set $H = \{\chi_m : m \in \mathbb{N}\}$ is not relatively weakly compact in $M(K)$. Indeed, otherwise Eberlein–Šmulian would give a subsequence converging weakly in $M(K)$, hence converging pointwise on $C(K)$. By Lemma 2.2, every member of $C(K)$ is the restriction of a member of $C(X)$, so the same subsequence would be weakly Cauchy in $L(X)$, a contradiction.

Apply Theorem 2.4. There are pairwise disjoint open sets $U_n \subseteq K$, measures $\mu_n \in H$, and $\varepsilon > 0$ such that

$$|\mu_n(U_n)| > 3\varepsilon \quad (n \in \mathbb{N}).$$

By regularity, choose a compact set $F_n \subseteq U_n$ such that $|\mu_n|(U_n \setminus F_n) < \varepsilon$. Normality of K gives $g_n \in C(K, [0, 1])$ with $g_n = 1$ on F_n and $\text{supp}(g_n) \subseteq U_n$. Then

$$|\mu_n(g_n)| > 2\varepsilon.$$

The supports of the g_n are pairwise disjoint.

Since K is compact in the Tychonoff space μX , it is C -embedded. Extend each g_n to a function $u_n \in C(\mu X, [0, 1])$. Define

$$\tilde{h}_0 = u_0, \quad \tilde{h}_n = u_n \prod_{j < n} (1 - u_j) \quad (n \geq 1),$$

and let $h_n = \tilde{h}_n|_X$. Because the supports of the g_n are disjoint,

$$\tilde{h}_n|_K = g_n.$$

Moreover, for every $x \in \mu X$ and every N ,

$$\sum_{n=0}^N \tilde{h}_n(x) = 1 - \prod_{n=0}^N (1 - u_n(x)) \leq 1.$$

This proves (ii), and since each μ_n is supported on K ,

$$\sup_{\chi \in A} |\langle h_n, \chi \rangle| \geq |\langle h_n, \mu_n \rangle| = |\mu_n(g_n)| > 2\varepsilon.$$

Thus (iii) holds with $\eta = 2\varepsilon$.

It remains to prove (i). Let $\Lambda \in (C(X)_\beta)'$. By Lemma 2.3, there are a functionally bounded set D and $\nu \in M(K_D)$ such that

$$\Lambda(f) = \int_{K_D} \bar{f} d\nu.$$

Since the same telescoping estimate holds on all of μX , Tonelli's theorem gives

$$\begin{aligned} \sum_n |\Lambda(h_n)| &\leq \sum_n \int_{K_D} \tilde{h}_n d|\nu| \\ &= \int_{K_D} \sum_n \tilde{h}_n d|\nu| \leq \|\nu\|. \end{aligned}$$

Thus $(\Lambda(h_n)) \in \ell_1$ for every $\Lambda \in (C(X)_\beta)'$, which is precisely weak 1-summability. $\square$

5 The universal weak variants

Theorem 5.1 (Universality of wsV^*). *For every Tychonoff space X and every $1 \leq p \leq q \leq \infty$, the space $L(X)$ has $wsV_{(p,q)}^*$.*

Proof. Let A be a (p, q) -(V^*) set. It is bounded because it is also a (p, q) -(EV^*) set and every such set is bounded. Suppose that A is not weakly sequentially precompact. Apply Lemma 4.1. The resulting sequence (h_n) is weakly 1-summable, hence weakly p -summable, in $C(X)_\beta$. However,

$$\sup_{\chi \in A} |\langle h_n, \chi \rangle| \geq \eta \quad (n \in \mathbb{N}),$$

so this sequence of suprema belongs neither to ℓ_q when $q < \infty$ nor to c_0 when $q = \infty$. This contradicts the (p, q) -(V^*) property of A . $\square$

Theorem 5.2 (Universality of finite- q $wsEV^*$). *Let $1 \leq p \leq q < \infty$. Then $L(X)$ has $wsEV^*_{(p,q)}$ for every Tychonoff space X .*

Proof. Let A be a (p, q) -(EV^*) set and suppose that it is not weakly sequentially precompact. It is bounded, so Lemma 4.1 provides $\eta > 0$ and a weakly 1-summable sequence (h_n) satisfying

$$\sum_n h_n(x) \leq 1 \quad (x \in X), \quad \sup_{\chi \in A} |\langle h_n, \chi \rangle| \geq \eta.$$

Choose positive scalars $b_n \rightarrow 0$ with $(b_n) \notin \ell_q$, for instance $b_n = (n+1)^{-1/q}$, and put $v_n = b_n h_n$. The sequence (v_n) remains weakly 1-summable, hence weakly p -summable.

We claim that $\{v_n : n \in \mathbb{N}\}$ is equicontinuous in $L(X)'$. The map

$$\Phi : X \longrightarrow c_0, \quad \Phi(x) = (v_n(x))_n,$$

is continuous: each coordinate is continuous, while

$$\sup_{x \in X} |v_n(x)| \leq b_n \longrightarrow 0,$$

so the tails are uniformly small. By the universal property of $L(X)$, Φ extends to a continuous linear operator $T : L(X) \rightarrow c_0$. Since $v_n = T'(e'_n)$ and the coordinate functionals (e'_n) form an equicontinuous subset of c'_0 , the sequence (v_n) is equicontinuous.

On the other hand,

$$\sup_{\chi \in A} |\langle v_n, \chi \rangle| \geq \eta b_n,$$

and $(b_n) \notin \ell_q$. This contradicts the (p, q) -(EV^*) property of A . $\square$

The damping $b_n \rightarrow 0$ is exactly what creates equicontinuity. It also explains why the preceding argument cannot contradict an (p, ∞) -(EV^*) condition: the lower bound ηb_n necessarily tends to zero. The endpoint $q = \infty$ therefore requires a separate analysis.

6 The exceptional endpoint $wsEV^*_p$

Proposition 6.1 (At $q = \infty$, p -(EV^*) means bounded). *For every Tychonoff space X and every $p \in [1, \infty]$, a subset of $L(X)$ is a p -(EV^*) set if and only if it is bounded.*

Proof. Every p -(EV^*) set is bounded [6, Lemma 7.2]. Conversely, let $B \subseteq L(X)$ be bounded. Put

$$D = \text{supp}(B), \quad C = C_B < \infty.$$

Let (f_n) be an equicontinuous weakly p -summable sequence in $C(X)_\beta$. Weak p -summability implies

$$f_n(x) \longrightarrow 0 \quad (x \in X),$$

because evaluation at x is a continuous functional on $C(X)_\beta$. Equicontinuity says that

$$\chi \longmapsto \sup_n |\langle f_n, \chi \rangle|$$

is a continuous seminorm on $L(X)$. Hence

$$T : L(X) \longrightarrow \ell_\infty, \quad T(\chi) = (\langle f_n, \chi \rangle)_n,$$

is continuous. Its restriction to X takes values in c_0 , so

$$\Phi : X \longrightarrow c_0, \quad \Phi(x) = (f_n(x))_n,$$

is continuous.

The set D is functionally bounded. Therefore $\Phi(D)$ is functionally bounded in the metric space c_0 , and hence has compact closure Q in c_0 . Compact subsets of c_0 have uniformly vanishing coordinates, so

$$\sup_{x \in D} |f_n(x)| = \sup_{z \in \Phi(D)} |z_n| \leq \sup_{z \in Q} |z_n| \longrightarrow 0.$$

Consequently,

$$\sup_{\chi \in B} |\langle f_n, \chi \rangle| \leq C \sup_{x \in D} |f_n(x)| \longrightarrow 0.$$

Thus B is a p -(EV*) set. □

Corollary 6.2. *For every $p \in [1, \infty]$,*

$L(X)$ has $wsEV_p^$ $\iff$ every bounded sequence in $L(X)$ has a weakly Cauchy subsequence.*

In particular, the endpoint property is independent of p .

Proof. By Proposition 6.1, the p -(EV*) subsets of $L(X)$ are exactly its bounded subsets. □

We next translate Corollary 6.2 into a condition on X . For a countable functionally bounded set $S \subseteq X$, put

$$K_S = cl_{\mu_X} S$$

and let

$$P_\omega(S) = \left\{ \sum_{j=1}^m t_j \delta_{s_j} : s_j \in S, t_j \geq 0, \sum_{j=1}^m t_j = 1 \right\} \subseteq P(K_S).$$

Here $P(K_S)$ is the weak-star compact space of Radon probability measures on K_S .

Definition 6.3 (Atomic sequential compactness property). *A Tychonoff space X has property ASCP if, for every countable functionally bounded set $S \subseteq X$, every sequence in $P_\omega(S)$ has a weak-star convergent subsequence in $P(K_S)$.*

Lemma 6.4 (Probability measures suffice). *Fix a countable functionally bounded $S \subseteq X$. The following are equivalent:*

- (i) *every sequence in $P_\omega(S)$ has a weak-star convergent subsequence in $P(K_S)$;*
- (ii) *every norm-bounded sequence of finitely supported real signed measures carried by S has a weak-star convergent subsequence in $M(K_S)$;*
- (iii) *every norm-bounded sequence of finitely supported complex measures carried by S has a weak-star convergent subsequence in $M(K_S)$.*

Proof. Only (i) $\Rightarrow$ (ii) requires an argument. Let (μ_n) be uniformly variation bounded and write the Jordan decomposition

$$\mu_n = \mu_n^+ - \mu_n^-.$$

After passing to a subsequence, the masses $\alpha_n = \mu_n^+(K_S)$ and $\beta_n = \mu_n^-(K_S)$ converge. If the limit of α_n is zero, then $\|\mu_n^+\| \rightarrow 0$; otherwise, normalize $\mu_n^+/\alpha_n \in P_\omega(S)$ and use (i). Treat the negative parts in the same way and take a diagonal subsequence. This proves (ii). Splitting a complex measure into its real and imaginary parts gives (iii). The reverse implications are immediate. □

Theorem 6.5 (Topological-measure characterization of the endpoint). *For a Tychonoff space X , the following are equivalent:*

- (i) $L(X)$ has wsEV_p^* for one, equivalently every, $p \in [1, \infty]$;
- (ii) every bounded sequence in $L(X)$ has a weakly Cauchy subsequence;
- (iii) X has ASCP.

Proof. The equivalence (i) $\Leftrightarrow$ (ii) is Corollary 6.2.

Assume (ii), let S be countable and functionally bounded, and let $(\mu_n) \subseteq P_\omega(S)$. By Proposition 2.1, (μ_n) is a bounded sequence in $L(X)$. It therefore has a weakly Cauchy subsequence. Every $g \in C(K_S)$ extends to a member of $C(X)$ by Lemma 2.2; hence this subsequence is weak-star Cauchy on $C(K_S)$. Its pointwise limit is a positive norm-one functional on $C(K_S)$, so it is a member of $P(K_S)$. Thus X has ASCP.

Conversely, assume ASCP and let (χ_n) be a bounded sequence in $L(X)$. Set

$$S = \bigcup_n \text{supp}(\chi_n).$$

Then S is countable and functionally bounded, and the variations $\|\chi_n\|_1$ are uniformly bounded. By Lemma 6.4, a subsequence converges weak-star in $M(K_S)$. Therefore its values converge on every restriction $R_S f$, $f \in C(X)$, so the same subsequence is weakly Cauchy in $L(X)$. This proves (ii). $\square$

The restriction-space formulation from the earlier partial result becomes especially transparent here.

Proposition 6.6 (Restriction-space form). *For every countable functionally bounded $S \subseteq X$, the space*

$$E_S = \overline{\{f|_S : f \in C(X)\}}^{\|\cdot\|_\infty} \subseteq \ell_\infty(S)$$

is canonically isometric to $C(K_S)$. In fact the displayed restriction range is already closed. Property ASCP says exactly that

$$\text{co}\{\delta_s : s \in S\} \subseteq E'_S$$

is relatively weak-star sequentially compact for every such S .

Proof. By Lemma 2.2, restriction from $C(X)$ to S has exactly the same range as restriction from $C(K_S)$ to its dense subset S . The latter map is an isometry, so its range is complete and therefore closed. $\square$

7 Consequences and examples

Corollary 7.1. *The endpoint wsEV_p^* holds, and hence all weak variants hold, under any of the following conditions:*

- (i) *for every countable functionally bounded $S \subseteq X$, the compactum K_S is metrizable;*
- (ii) *for every such S , the compactum K_S is Rosenthal compact;*
- (iii) *X is metrizable.*

Proof. If K_S is metrizable, then $P(K_S)$ is compact metrizable and therefore sequentially compact. If K_S is Rosenthal compact, then $P(K_S)$ is Rosenthal compact by Godefroy's theorem [3]; Rosenthal compacta are Fréchet–Urysohn by the Bourgain–Fremlin–Talagrand theorem [1], and compact Fréchet–Urysohn spaces are sequentially compact. In either case ASCP follows.

Every metrizable space is a μ -space: functionally bounded subsets have compact closure. Hence, for countable functionally bounded S in a metrizable X , the compactum $K_S = \text{cl}_X S$ is metrizable. $\square$

Example 7.2 (Separation from the four stronger variants). *If $X = [0, 1]$, then $L(X)$ has every $wsV_{(p,q)}^*$ and every $wsEV_{(p,q)}^*$, but it has none of*

$$V_{(p,q)}^*, \quad EV_{(p,q)}^*, \quad sV_{(p,q)}^*, \quad sEV_{(p,q)}^*.$$

Indeed, the weak properties follow from Corollary 7.1, while the four stronger properties fail by Theorem 3.3 because $[0, 1]$ is an infinite functionally bounded set.

Example 7.3 (The endpoint is not universal). *Let $X = \beta\mathbb{N}$. The sequence (δ_n) is bounded in $L(\beta\mathbb{N})$, but it has no weakly Cauchy subsequence. Indeed, given a subsequence (n_k) , choose $a \in \ell_\infty = C(\beta\mathbb{N})$ with $a(n_k) = (-1)^k$. Then $\langle a, \delta_{n_k} \rangle = (-1)^k$ is not Cauchy. Thus $\beta\mathbb{N}$ fails ASCP, and*

$$L(\beta\mathbb{N}) \text{ fails } wsEV_p^* \text{ for every } p.$$

Nevertheless, Theorems 5.1 and 5.2 show that $L(\beta\mathbb{N})$ has all $wsV_{(p,q)}^$ properties and all $wsEV_{(p,q)}^*$ properties with $q < \infty$.*

Remark 7.4 (Why the endpoint does not collapse to a standard adjective). *The condition ASCP is an exact condition on X , but it need not reduce to metrizability of the compacta K_S . Sequential properties of spaces of probability measures over nonmetrizable compacta are delicate. For example, it remains open in general whether $P(K)$ must be sequentially compact whenever K is sequentially compact; see [7, Problem 7.8(4)]. Thus an attempt to replace ASCP by a familiar one-word compactness condition would encounter a genuinely nontrivial problem in the topology of $P(K)$ spaces.*

8 Conclusion

Theorem 1.1 follows by combining Theorems 3.3, 5.1, 5.2, and 6.5. In compact form, the answer is

Property of $L(X)$	Exact condition on X
$V_{(p,q)}^*, EV_{(p,q)}^*, sV_{(p,q)}^*, sEV_{(p,q)}^*$	Every functionally bounded subset of X is finite
$wsV_{(p,q)}^*$	Always
$wsEV_{(p,q)}^*, q < \infty$	Always
$wsEV_p^*$	ASCP

The central new mechanism is Lemma 4.1: the Dieudonné–Grothendieck disjoint-open-set criterion is combined with a telescoping extension to turn failure of weak sequential precompactness into a weakly 1-summable detector in the strong dual of $L(X)$.

Verification status. The weighted-partial-sum argument for the four stronger variants was already present in the supplied partial packet. The detector lemma and its consequences for the weak variants are the additional part of this proof draft. The argument has been written so that all nonstandard steps are isolated in Lemmas 2.2, 2.3, and 4.1. The local run registry was checked for prior packets on arXiv:2406.07178, and a targeted web/arXiv search on June 18, 2026 for Problem 3.18, arXiv:2406.07178, and the phrases “ V^* ”, “ $wsEV^*$ ”, and “free locally convex” located the source problem and the background Pełczyński V^* paper but no separate published solution of Problem 3.18. This is a bounded novelty check, not a formal exhaustive review. Human review should pay particular attention to the detector lemma, the use of the Dieudonné–Grothendieck criterion, and the endpoint identification of p -(EV^*) sets with bounded sets.

References

- [1] J. Bourgain, D. H. Fremlin, and M. Talagrand, Pointwise compact sets of measurable functions, *Amer. J. Math.* **100** (1978), 845–886.
- [2] T. Banach and S. Gabrielyan, On free locally convex spaces, *Filomat* **36** (2022), no. 18, 6393–6401.
- [3] G. Godefroy, Compacts de Rosenthal, *Pacific J. Math.* **91** (1980), 293–306.
- [4] S. Gabrielyan, Locally convex properties of free locally convex spaces, *J. Math. Anal. Appl.* **480** (2019), 123453.
- [5] S. Gabrielyan, Gelfand–Phillips type properties of locally convex spaces, *Mathematics* **12** (2024), no. 22, 3537.
- [6] S. Gabrielyan, Pełczyński’s type sets and Pełczyński’s geometrical properties of locally convex spaces, arXiv:2402.08860 (2024).
- [7] G. Plebanek, A survey on topological properties of $P(K)$ spaces, *Collect. Math.* (2024), doi:10.1007/s11537-024-2410-y.
- [8] B. B. Wells, Jr., Weak compactness of measures, *Proc. Amer. Math. Soc.* **20** (1969), 124–130.